Module 02 – Extra Class

BASIC PROBABILITY

Nguyen Quoc Thai

Objectives

Introduction

- Experiment
- Even
- Operations on Events
- Relations of events

Probability

- Definition
- Rule of Probability
- Total Probability
- Bayes' Rule

Outline

SECTION 1
Introduction

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SECTION 3
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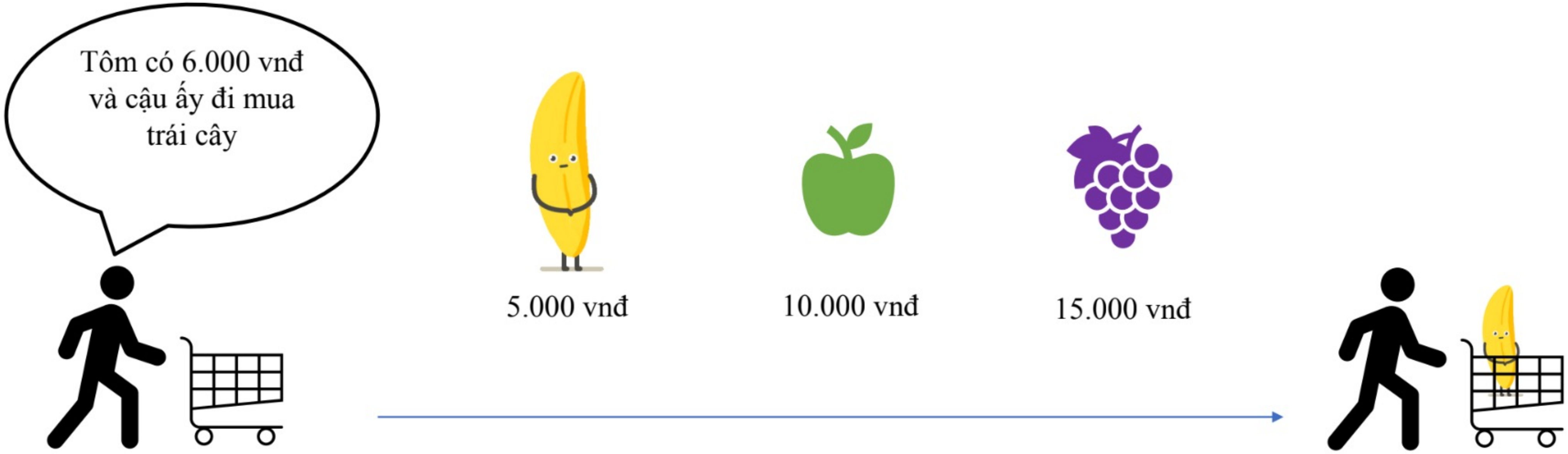
SECTION 4
Bayes' Rule

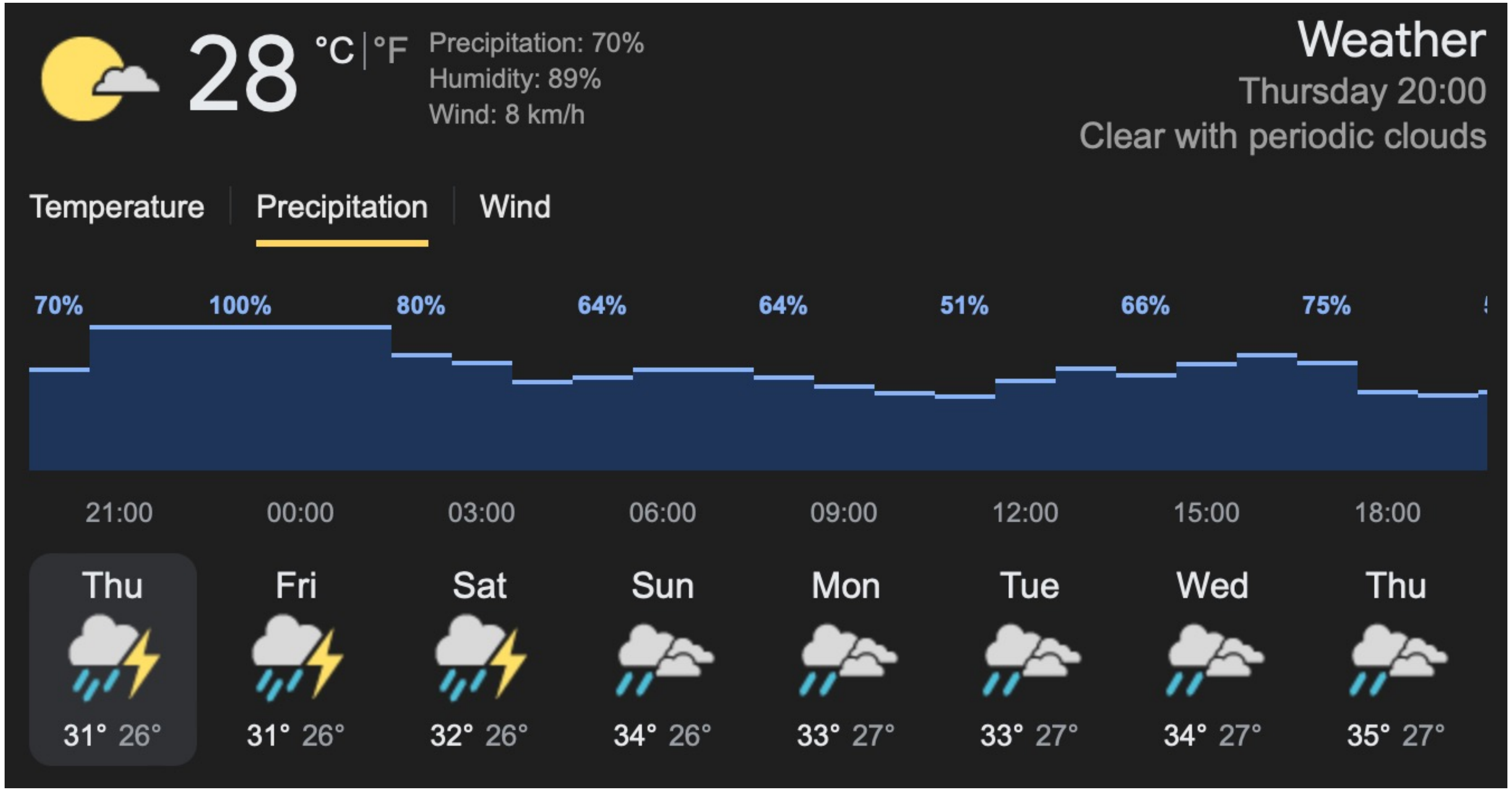
Introduction

Probability



Introduction

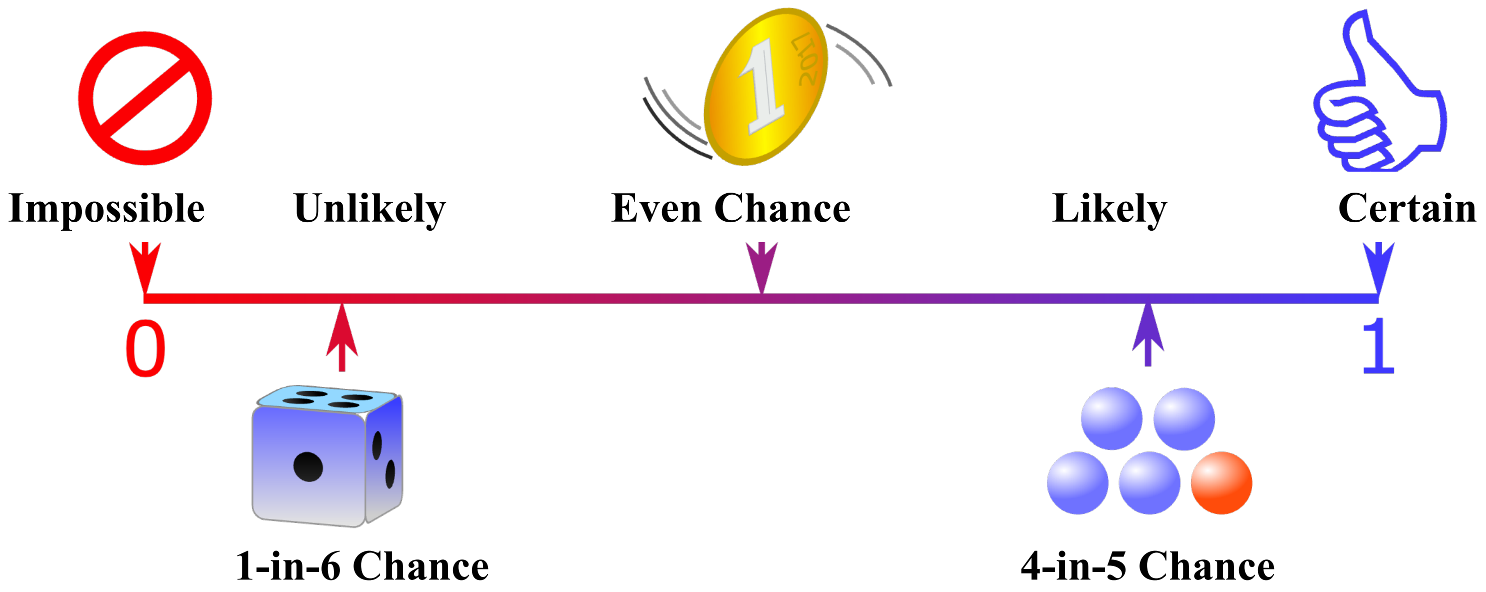
Probability



Introduction

Probability

- Measure to the likelihood of an event occurring



Introduction

Experiment & Event

- Experiment: implementation of set of basic conditions for observing a certain phenomenon
- An outcome: a result of an experiment
- A sample space: the set of all possible outcomes
- An event: a subset of the sample space

Introduction

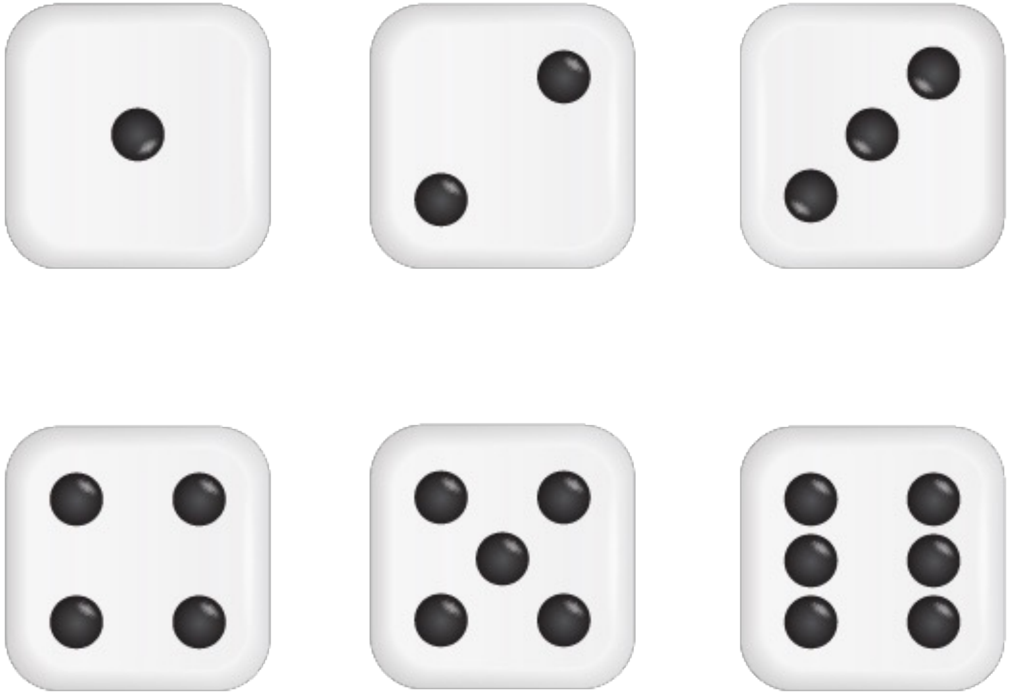
Experiment & Event

Toss a coin

Sample space: S = {*heads*, *tails*}

Roll a dice

Sample space: S = {1, 2, 3, 4, 5, 6}



Introduction

Relation of events

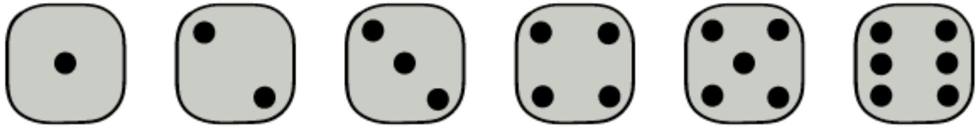
Assume A and B: two events in the same experiment

- Implication
  - “event A implies event B”: if event A occurs, then event B occurs
  - $A \Rightarrow B$ means that $A \subseteq B$
- Equivalent
  - “event A equal event B”: if $A \Rightarrow B$ and $B \Rightarrow A$
  - $A \Leftrightarrow B$

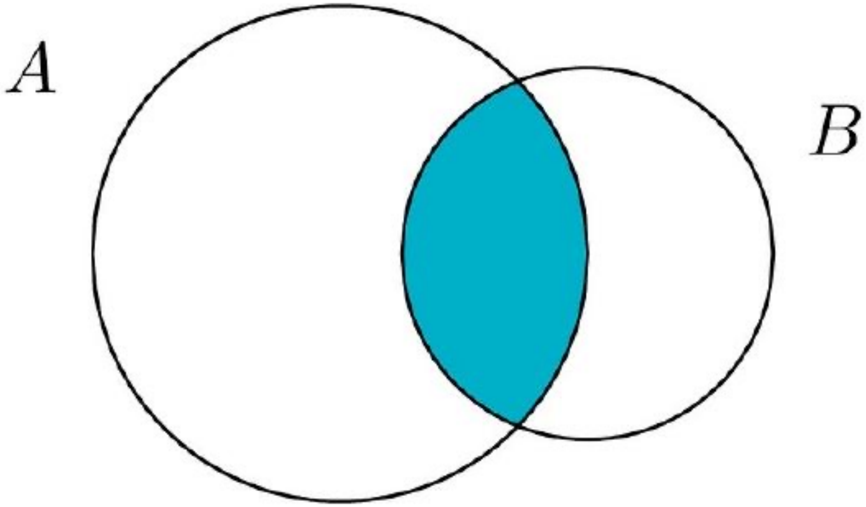
Introduction

Operations on Events

- **Intersection of events (A ∩ B)**
  - In the experiment of rolling a single die



  - Event A: "the number rolled is even"
  - Event B: "the number rolled is divisible by 3"
  - Find the intersection of A and B?



Introduction

Operations on Events

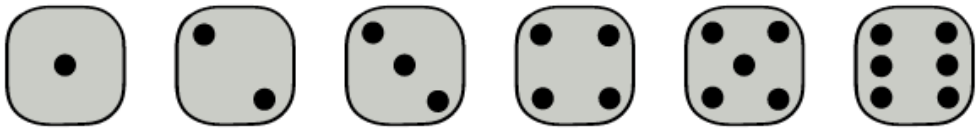
- **Intersection of events (A ∩ B)**
  - In the experiment of rolling a single die
  - Event A: “the number rolled is even”

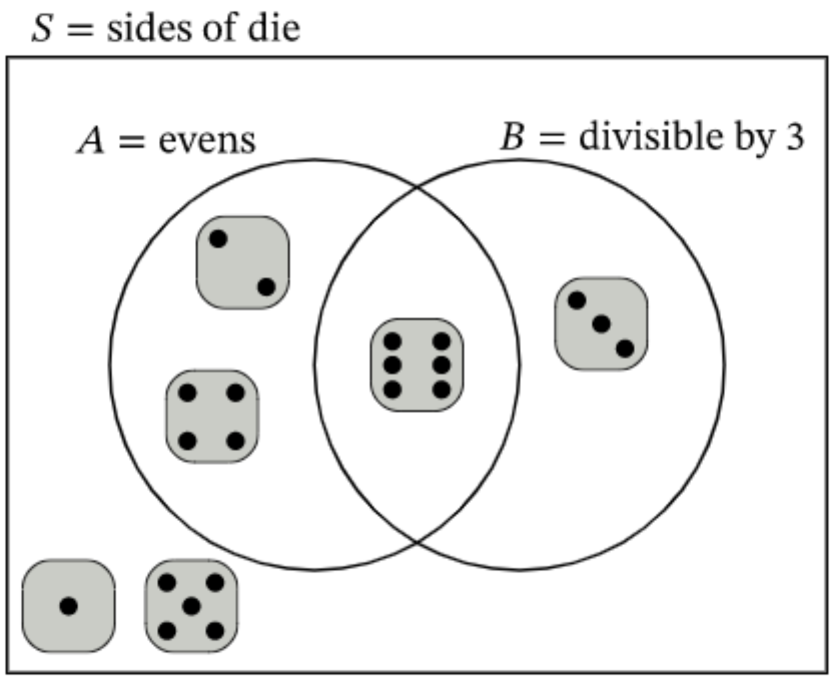
=> A = {2, 4, 6}

  - Event B: “the number rolled is divisible by 3”

=> B = {3, 6 }

  - Find the intersection of A and B?





Introduction

Operations on Events

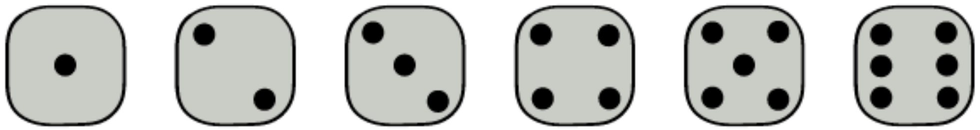
- **Intersection of events ($A \cap B$)**
  - In the experiment of rolling a single die
  - Event A: “the number rolled is even”

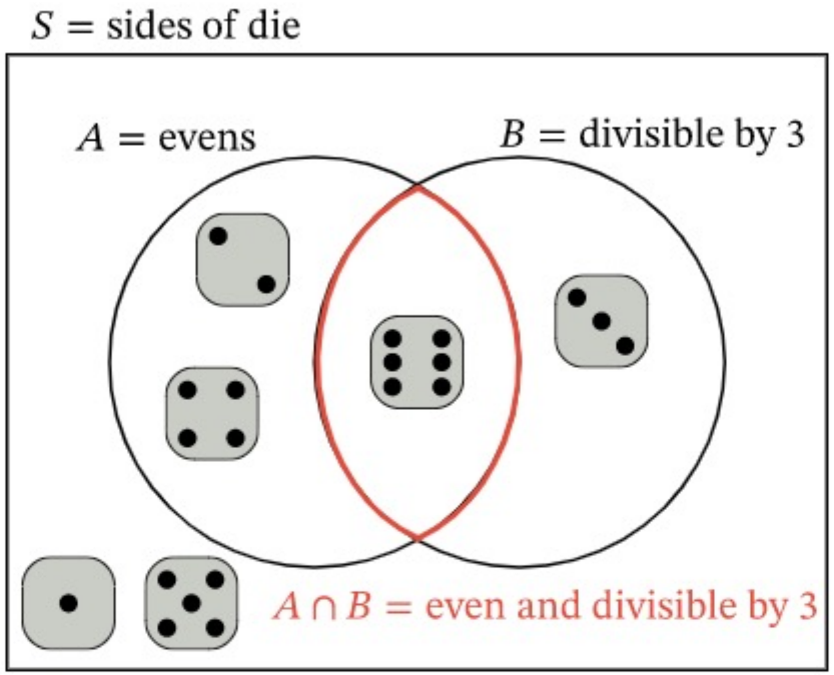
=> $A = \{2, 4, 6\}$

  - Event B: “the number rolled is divisible by 3”

=> $B = \{3, 6\}$

  - $A \cap B = \{2, 3, 4, 6\}$

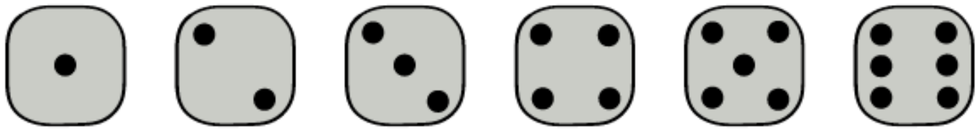


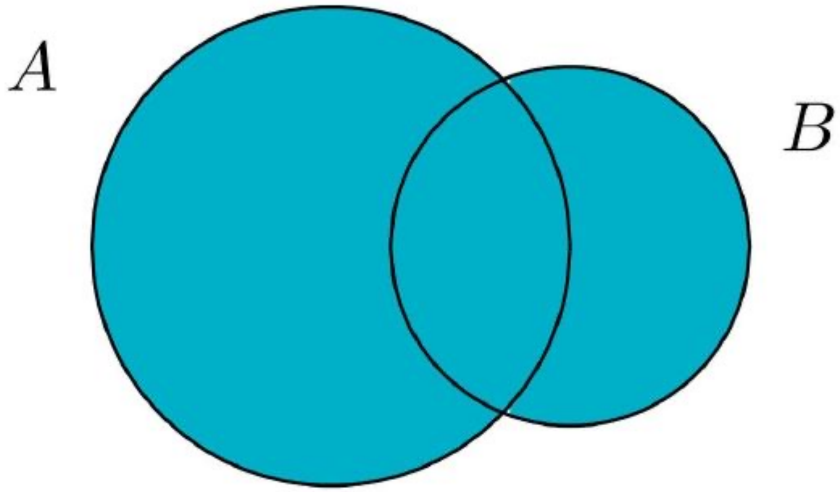


Introduction

Operations on Events

- **Union of events (A ∪ B)**
  - In the experiment of rolling a single die
  - Event A: “the number rolled is even”
  - Event B: “the number rolled is divisible by 3”
  - Find the union of A and B?





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Introduction

Operations on Events

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  - Event A: “the number rolled is even”

=> A = {2, 4, 6}

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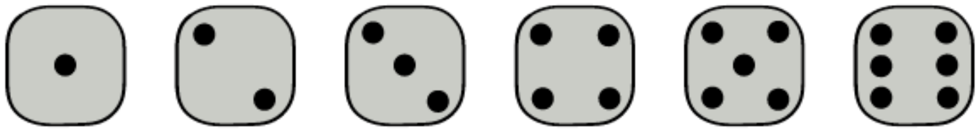

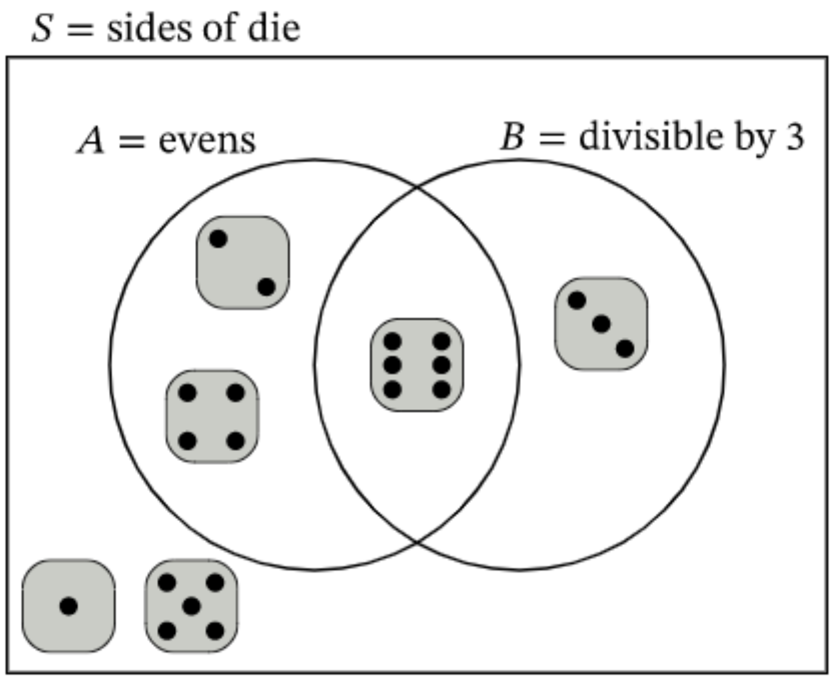

Introduction

Operations on Events

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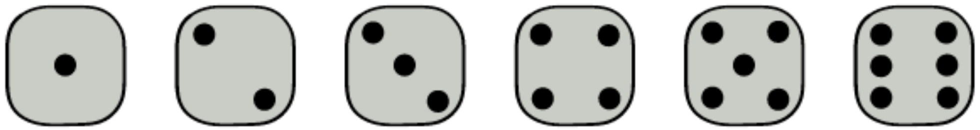
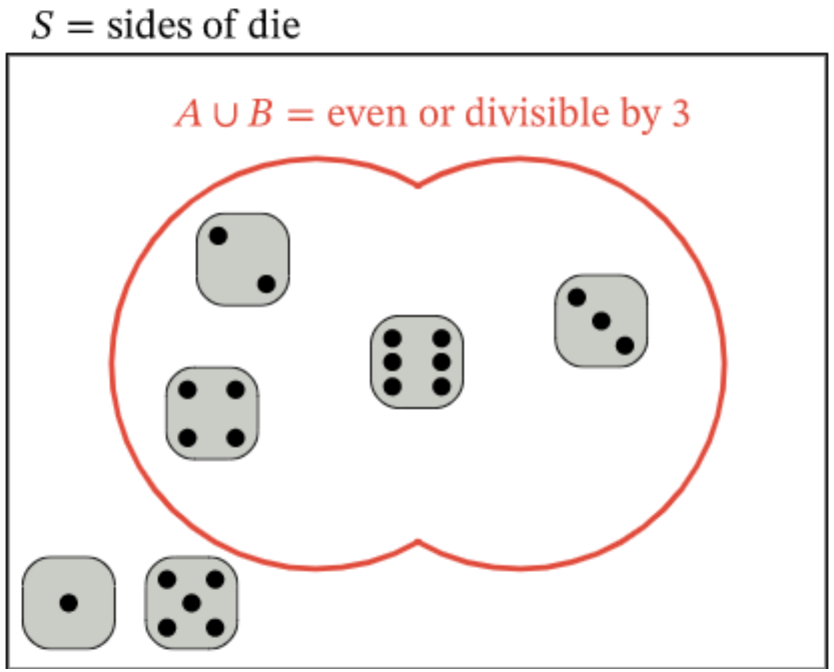
  - Event B: “the number rolled is divisible by 3”

=> B = {3, 6 }

  - A ∪ B = {2, 3, 4, 6}



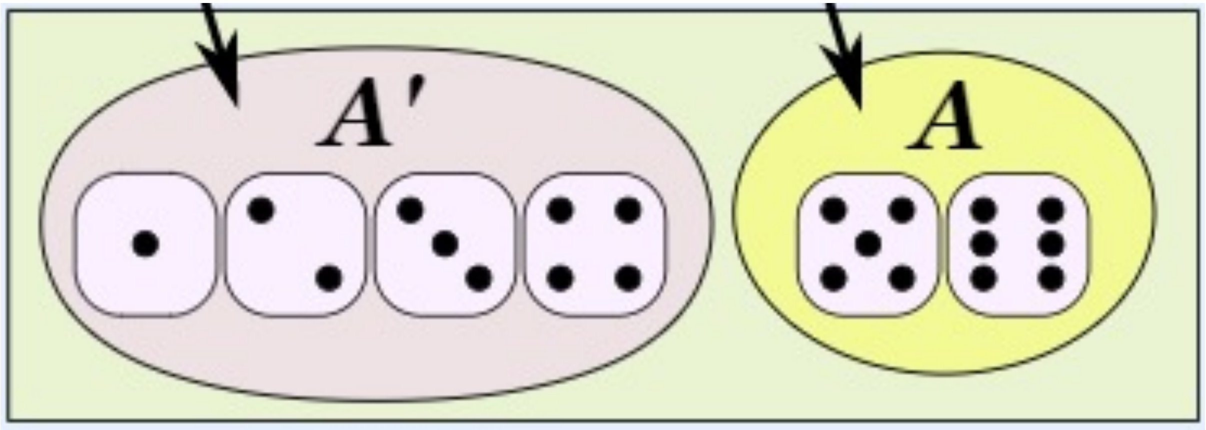


Introduction

Operations on Events

- **Complements**
  - The complement of an event A in a sample space S, denoted A’ (A^c)
  - The collection of all outcomes in S that are not elements of the set A
  - Corresponds to negating any description in words of the event A.
  - $A' + A = \Omega$

Complement of an event A An event A



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Probability

Classical Probability

$$P(A) = \frac{number\ of\ favorable\ outcomes}{total\ number\ of\ possible\ outcomes} = \frac{n_A}{n_\Omega}$$

Example

What is the probability of rolling a number is even on a regular dice?

- There are 6 faces on a fair die, numbered 1 to 6 => n(Ω) = 6
- A : "even number" => A = {2, 4, 6} => n(A) = 3

=> P(A) = 3/6 = 0.5

Probability

Geometric Probability

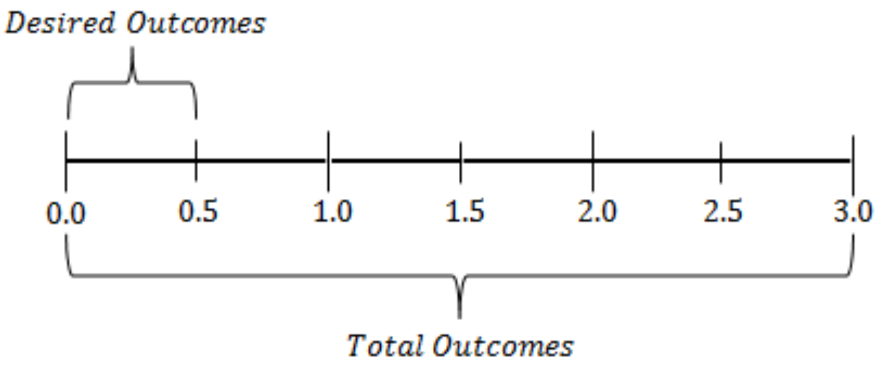
$$P(A) = \frac{measure\ of\ domain\ A}{measure\ of\ domain\ \Omega}$$

1-D Geometric probability

X is a random real number between 0 and 3. What is the probability X is closer to 0 than it is to 1?

=> A: "X is closer to 0 than to 1"

=> Measure: length in this 1D case: $P(A) = \frac{length\ of\ segment\ where\ 0<X<0.5}{length\ of\ segment\ where\ 0<X<3} = \frac{0.5}{3} = \frac{1}{6}$



Probability

Geometric Probability

$$P(A) = \frac{measure\ of\ domain\ A}{measure\ of\ domain\ \Omega}$$

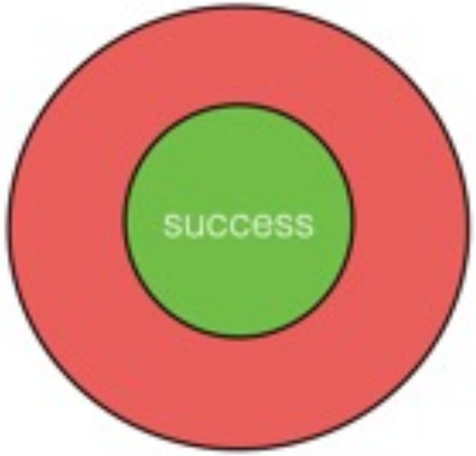
2-D Geometric probability

A dart is thrown at a circular dartboard such that it will land randomly over the area of the dartboard. What is the probability that it lands closer to the center "success" than to the edge?

=> A: "closer to center than edge"

=> Measure: area in this 2D case:

$$P(A) = \frac{area\ of\ desired\ outcomes}{area\ of\ total\ outcomes} = \frac{\frac{\pi r^2}{4}}{\pi r^2} = \frac{1}{4}$$



Probability

Rules of Probability

Addition

- Mutually exclusive events

$$P(A+B) = P(A) + P(B), \text{ where A and B mutually exclusive}$$

$$P(A \text{ or } B) = P(A) + P(B)$$

- In general

$$P(A+B) = P(A) + P(B) - P(AB)$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Probability

Rules of Probability

Addition

Rolling a fair die. What is the probability of $A = \{1, 5\}$?

- The problem states that the die is fair => all six possible outcomes are equally likely:

$$P(\{1\}) = P(\{2\}) = P(\{3\}) = P(\{4\}) = P(\{5\}) = P(\{6\})$$

- The events $\{1\},..\{6\}$ are disjoint:

$$1 = P(S) = P(\{1\}) + P(\{2\}) + \ldots + P(\{6\}) = 6P(\{1\})$$

=> $P(\{1\}) = P(\{2\}) = P(\{3\}) = P(\{4\}) = P(\{5\}) = P(\{6\}) = 1/6$

- Since $\{1\}$ and $\{5\}$ are disjoint:

=> $P(A) = P(\{1, 5\}) = P(\{1\}) + P(\{5\}) = 2/6 = 1/3$

Probability

Rules of Probability

Addition

- For any event A

$$P(A^c) = 1 – P(A)$$

$$P(A) = 1 – P(A^c)$$

Find the probability that when we roll a dice we get a number different than 1 and 6?

- Let's A: "Getting the number 1 and 6" => A = {1, 6}

  "Getting a number different than 1 and 6" = A^c

  Since, P(A) = P(1) + P(6) = 1/6 + 1/6 = 2/6 = 1/3

  P("Getting a number different than 1 and 6") = 1 – P(A) = 1 – 1/3 = 2/3

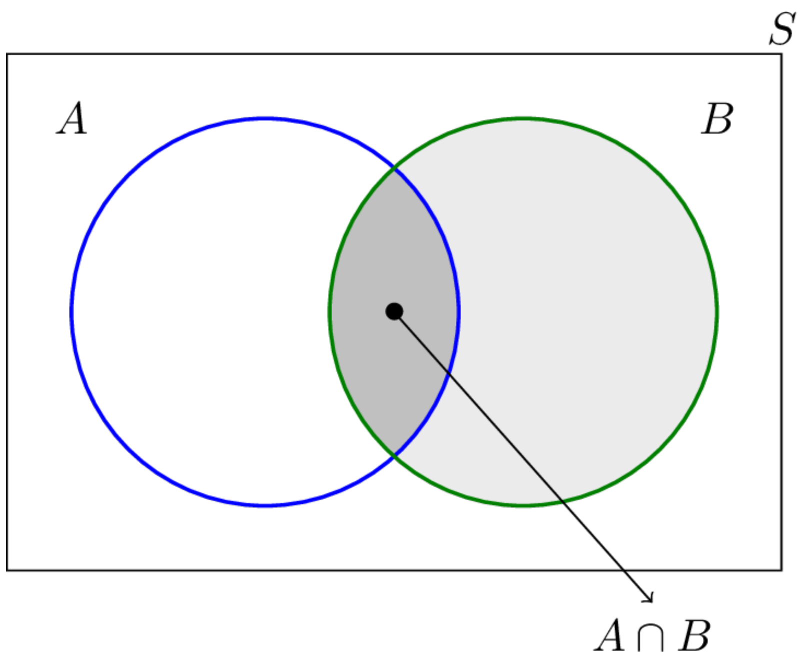
Probability

Rules of Probability

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability that A occurs given that B has already occurred



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability

Rules of Probability

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

A fair die is rolled

a) Find the probability that the number rolled is a five, given that it is odd.

b) Find the probability that the number rolled is odd, given that it is a five.

Probability

Rules of Probability

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

A fair die is rolled

- Sample space S = {1, 2, 3, 4, 5, 6}, consisting of 6 equally likely outcomes
- A: "a five is rolled" => A = {5} => P(A) = 1/6
- B: "an odd number is rolled" => B = {1, 3, 5} => P(B) = 3/6 = ½

=> A and B = {5} => P(A and B) = 1/6

a) Find the probability that the number rolled is a five, given that it is odd.

P(A|B) = P(A and B)/P(B) = (1/6)/(1/2) = 1/3

Probability

Rules of Probability

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

A fair die is rolled

- Sample space S = {1, 2, 3, 4, 5, 6}, consisting of 6 equally likely outcomes
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=> A and B = {5} => P(A and B) = 1/6

a) Find the probability that the number rolled is odd, given that it is a five.

P(B|A) = P(B and A)/P(A) = P(A and B)/P(A) = (1/6)/(1/6) = 1

Probability

Rules of Probability

Multiplication

$$P(AB) = P(A).P(B|A) = P(B).P(A|B)$$

$$P(A_1A_2\ldots A_n) = P(A_1).P(A_2|A_1).P(A_3|A_1A_2)\ldots P(A_n|A_1A_2\ldots A_{n-1})$$

- In a factory there are 100 units of a certain product, 5 of which are defective. We pick three units from the 100 units at random. What is the probability that none of them are defective?

Probability

Rules of Probability

Multiplication

Let's A_i as the event i^{th} chosen unit is not defective, for i =1, 2, 3

=> Compute $P(A_1A_2A_3)$

- $P(A_1) = 95/100$
- Given that the first chosen item was good, the second item will be chosen from 94 good units and 5 defective units, thus: $P(A_2|A_1) = 94/99$
- Given that the first and second chosen items were okay, the third item will be chosen from 93 good units and 5 defective units, thus: $P(A_3|A_2A_1) = 93/98$

=> $P(A_1A_2A_3) = 95/100.94/99.93/98 = 0.8560$

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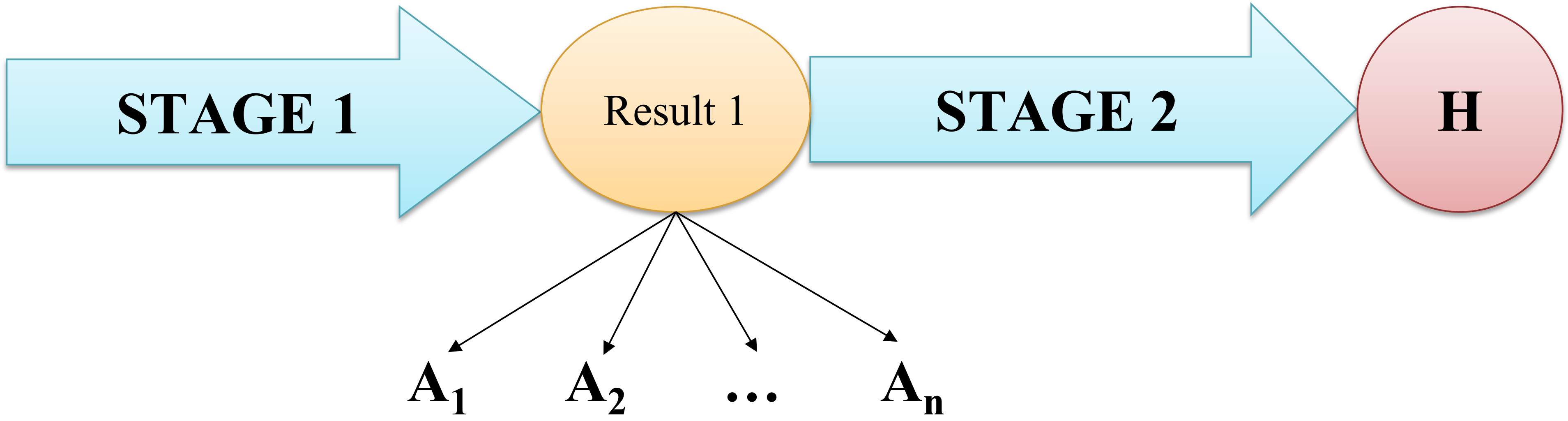
Total Probability Theorem

Problem

- The result of stage 2 depends on the result of stage 1. The results of step 1 are divided into n sets A_i, each of which will contain a number of outcomes that have the same effect on the probability of H occurring.

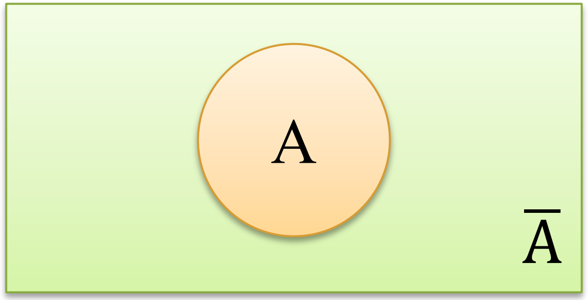


Total Probability Theorem

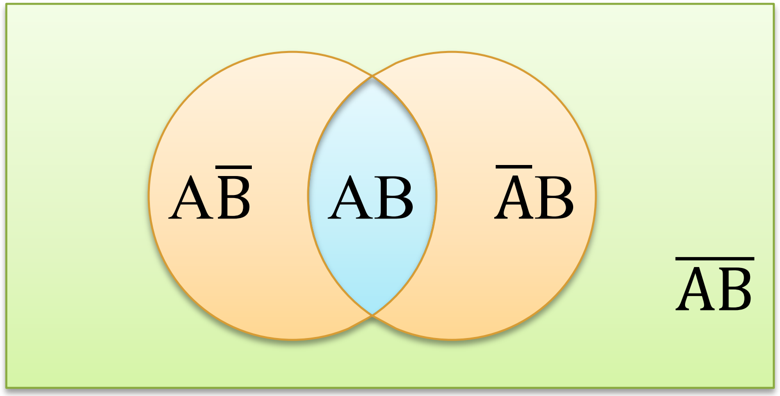
Problem

- Complete system of events
- Events A1, A2,…, An (n 2) of a trial is called complete system if:
  - $A_i \,.\, A_j = \emptyset, \forall\, i \neq j$
  - $\sum_{i=1}^{n} A_i = A_1 + A_2 + A_3 + \cdots + A_n = \Omega$
  - $P(A_1) + P(A_2) + \ldots + P(A_n) = 1$

$\{A, \bar{A}\}$



$\{AB, \bar{A}B, A\bar{B}, \overline{AB}\}$

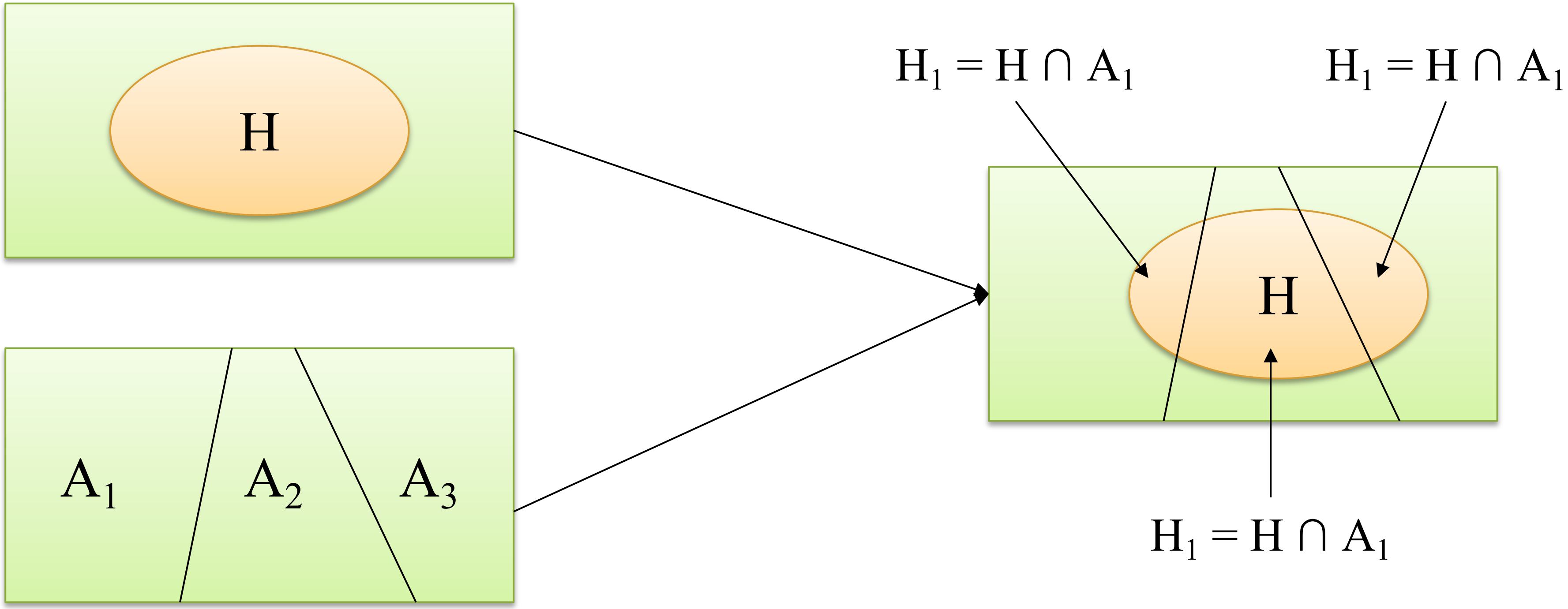


Total Probability Theorem

Total Probability

- Let A_1, A_2, …, An – complete system of events and assume. Consider any event H such that H occurs only when one of the events $A_1, A_2, \ldots, A_n$ occurred:

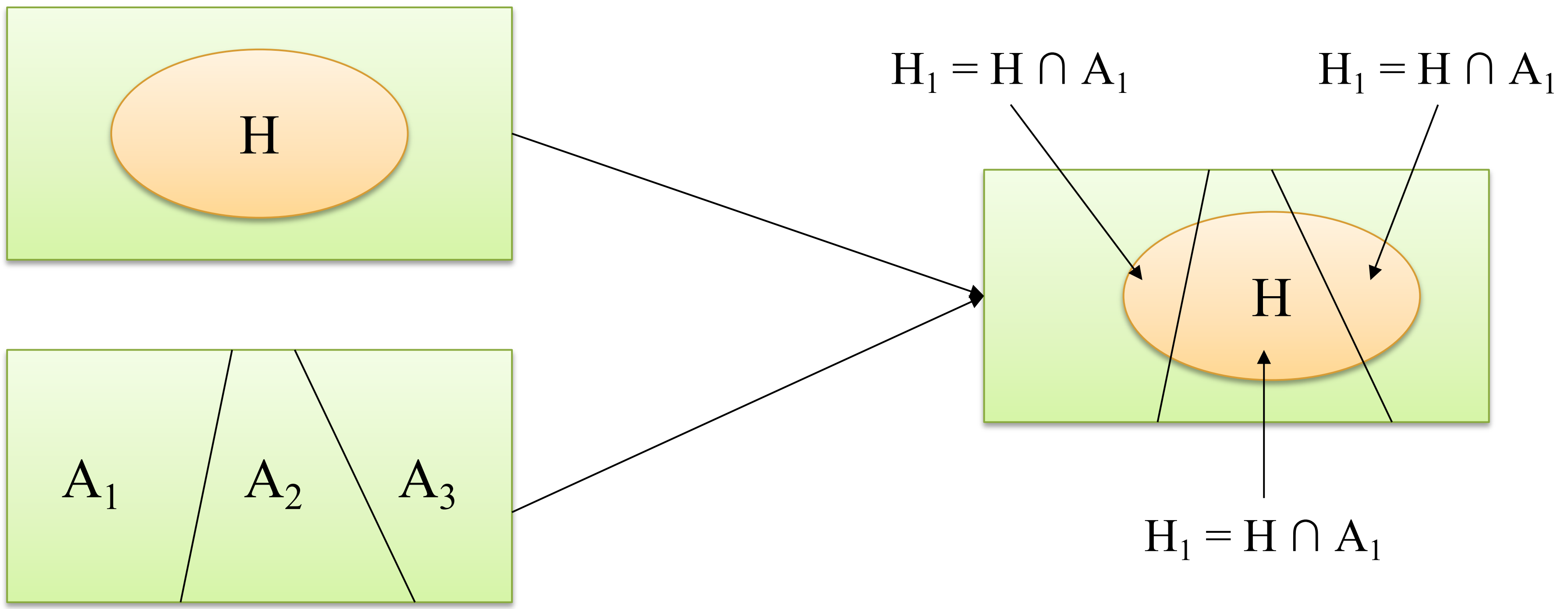
$$P(H) = P(H_1) + P(H_2) + P(H_3)$$
$$= P(A_1).P(H|A_1) + P(A_2).P(H|A_2) + P(A_3).P(H|A_3)$$



Total Probability Theorem

Total Probability

$$P(H) = \sum_{i=1}^{n} P(A_i).P(H|A_i)$$



Total Probability Theorem

Total Probability

- Company M supplies 80% of widgets for a car shop and only 1% of their widgets turn out to be defective. Company N supplies the remaining 20% of widgets for the car shop and 3% of their widgets turn out to be defective. If a customer randomly purchases a widget from the car shop, what is the probability that it will be defective?

Total Probability Theorem

Total Probability

$$P(H) = \sum_{i=1}^{n} P(A_i).P(H|A_i)$$

Company M	Supplies 80% of widgets 1% are defective
Company N	Supplies 20% of widgets 3% are defective

H: “Widget being defective”

A_M: “Widget came from company M”, A_N: “Widget came from company N”

Events A_M and A_N : complete system of events

=> $P(A_M) = 0.8$; $P(A_N) = 0.2$; $P(H|A_M) = 0.01$; $P(H|A_N) = 0.03$

The probability that it will be defective:

$P(H) = P(H|A_M).\ P(A_M) + P(H|A_N).\ P(A_N)$

$= 0.01*0.8 + 0.03*0.2$

$= 0.014$

QUIZ TIME

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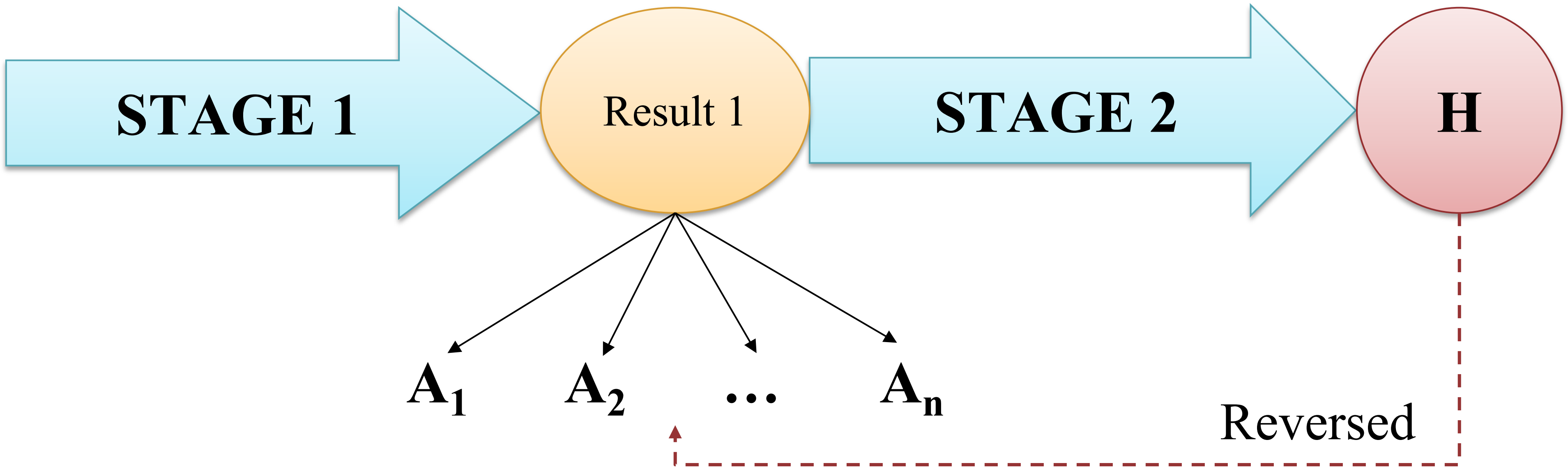
Bayes' Rule

Objective Bayes' Rule

- Based on the outcome H, calculate the probability that the i^{th} cause occur: $P(A_i|H)$



Bayes' Rule

Objective Bayes' Rule

LIKELIHOOD
The probability of "A" being True. Given "B" True

PRIOR
The probability of "B" being True. This is the knowledge

$$P(B|A) = \frac{P(A|B).P(B)}{P(A)}$$

POSTERIOR
The probability of "B" being True. Given "A" True

MARGINALIZATION
The probability of "A" being True.

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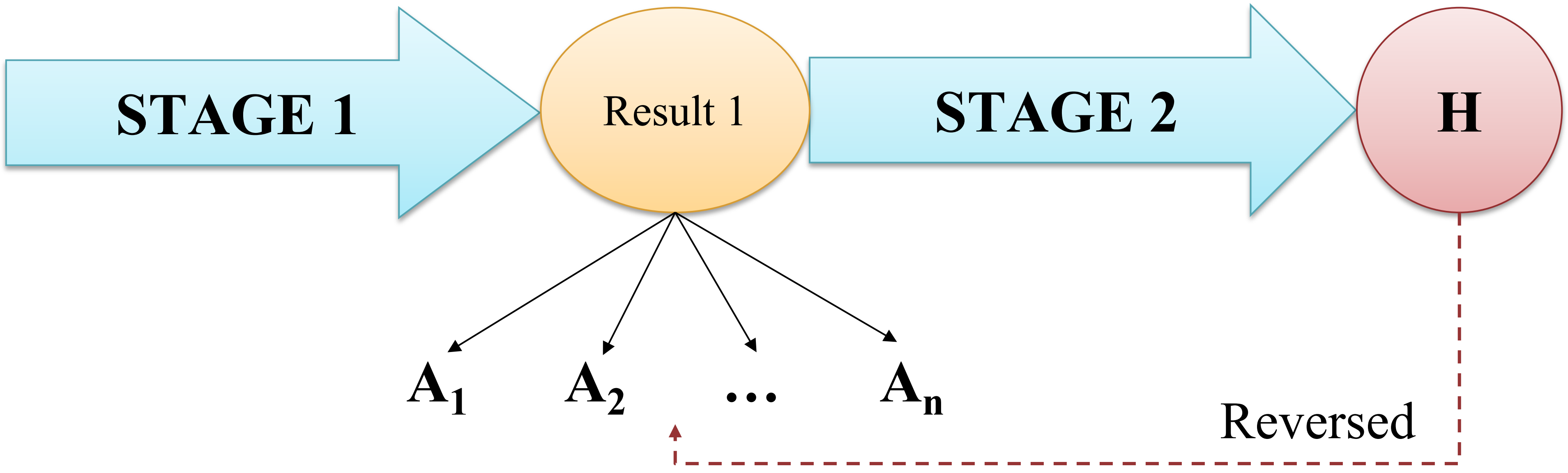
Bayes' Rule

Objective Bayes' Rule

- If $A_1, A_2, .. A_n$: complete system of events and H is any event with $P(A) \neq 0$:

$$P(A_i|H) = \frac{P(A_i)P(H|A_i)}{P(H)} = \frac{P(A_i)P(H|A_i)}{\sum_{j=1}^{n} P(A_j)P(H|A_j)}, i = 1, 2, \ldots, n$$

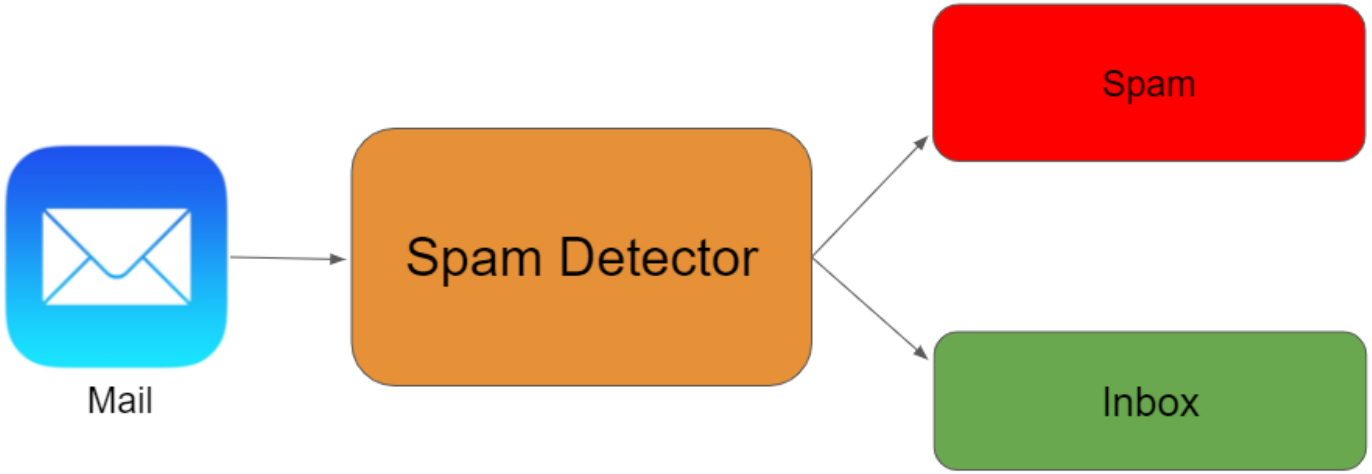


Bayes' Rule

Objective Bayes' Rule

Detect Spam E-Mail (Simple NLP problem)

Assume that the word 'offer' occurs in 80% of the spam messages in my account. Also, let's assume 'offer' occurs in 10% of my desired e-mails. If 30% of the received e-mails are considered as a scam, and I will receive a new message which contains 'offer', what is the probability that it is spam?



Bayes' Rule

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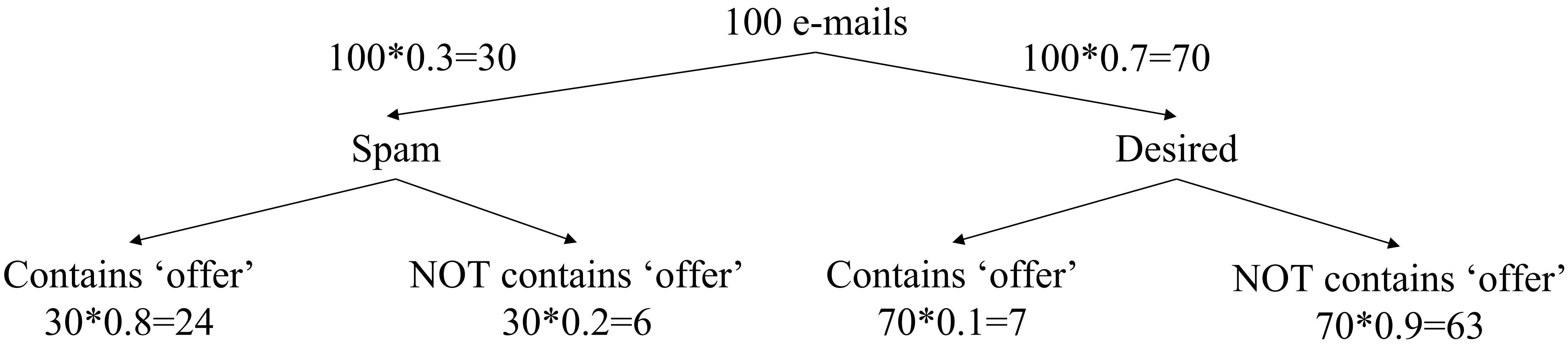

Bayes' Rule

Objective Bayes' Rule

Detect Spam E-Mail (Simple NLP problem)

Let A_1: "Spam", A_2: "Not spam" => A_1, A_2 : complete system of events

H: "contains the word 'offer'"

If a new message which contains 'offer', the probability that it is spam is:

$$P(A_1|H) = \frac{P(A_1)P(H|A_1)}{P(H)}$$

$P(H|A_1) = 0.8$; $P(A_1) = 0.3$, $P(A_2) = 1 – P(A_1) = 0.7$, $P(H|A_2) = 0.1$

$P(H) = P(A_1).P(H|A_1) + P(A_2).P(H|A_2) = 0.3*0.8 + 0.7*0.1 = 0.31$

=> $P(A_1|H) = (0.8*0.3)/(0.31) = 0.774$

Summary

Introduction

- Experiment
- Even
- Operations on Events
  - Intersection
  - Union
  - Complement
- Relations of events

Probability

- Definition

$P(A) = \frac{n_A}{n_\Omega}$ $P(A) = \frac{S_A}{S}$

- Rule of Probability

Addition: P(A+B) = P(A) + P(B) – P(AB)

Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Multiplication:

P(AB) = P(A).P(B|A) = P(B).P(A|B)

- Total Probability

$$P(H) = \sum_{i=1}^{n} P(A_i).P(H|A_i)$$

- Bayes' Rule

$$P(A_i|H) = \frac{P(A_i)P(H|A_i)}{P(H)}$$

Thanks!

Any questions?